Horizon boundary mechanics

Area, acceleration length, and horizon work

J. T. Tyler La Jolla, CA, USA 12 October 2026

Area measures transverse size. Acceleration length is the light-travel length of the surface-gravity timescale, $L = ct_\kappa$ with $t_\kappa = c/\kappa$. One boundary potential gives surface stress, force, and pressure.

Geometry and the mechanical responses

For a specified clock and positive surface gravity κ, use the horizon area A and

$$L = \frac{c^2}{\kappa}, \qquad U(A, L) = \frac{A\kappa c^2}{8\pi G} = \frac{c^4}{8\pi G}\frac{A}{L}. \tag{1}$$

The Einstein boundary structure fixes this potential [1]. With c, G fixed,

$$dU = \frac{c^4}{8\pi GL}\, dA - \frac{Ac^4}{8\pi GL^2}\, dL = \Pi\, dA - F_L\, dL. \tag{2}$$

For the mass equivalent $M_U = U/c^2$, the responses and force density are

$$\Pi = \left(\frac{\partial U}{\partial A}\right)_L = \frac{U}{A}, \qquad F_L = -\left(\frac{\partial U}{\partial L}\right)_A = \frac{U}{L} = M_U\kappa, \tag{3}$$

$$P_L = \frac{F_L}{A} = \frac{\Pi}{L} = \frac{\kappa^2}{8\pi G}, \qquad U = \Pi A = F_L L = P_L AL. \tag{4}$$

The two terms change surface area and energy per area: $A\, d\Pi = -F_L\, dL$, or $d\Pi = -P_L\, dL$. Volume work uses the pressure paired with its displacement; AL is not generally that volume.

Thermal translation and stationary horizons

The Hawking–Unruh temperature and Einstein area entropy give

$$T = \frac{\hbar c}{2\pi k_B L}, \qquad S = \frac{k_B A}{4\ell_P^2}, \quad \ell_P^2 = \frac{\hbar G}{c^3}, \qquad U = TS, \quad \Pi\, dA = T\, dS. \tag{5}$$

Rindler: $L = c^2/a_{\text{obs}}$ is the normal horizon distance on the instantaneous rest slice; planar patch area A is independent. *Schwarzschild:* $A = 4\pi R^2$, $L = 2R$, $U = Mc^2/2$, with time normalized at infinity. For areal volume $V = 4\pi R^3/3$,

$$F_L\, dL = F_R\, dR = M\kappa\, dR = P_R\, dV, \qquad P_R = 2P_L = \frac{\kappa^2}{4\pi G}. \tag{6}$$

Since $dL/L = dR/R$, this describes the same fractional horizon-size change. R is areal radius, not a universal proper distance. Thus $dU = \Pi\, dA - P_R\, dV$, with $U = 3P_R V$; area and volume co-vary. This reveals the mass-energy decomposition $Mc^2 = TS + F_L L$, while retaining the standard mass first law $d(Mc^2) = T\, dS$.

Cosmological payoff

For the equilibrium FRW apparent horizon, $L = R_A$, so $P \equiv P_R = P_L$. With $V = 4\pi R_A^3/3$, the Einstein–Misner–Sharp identification gives [2]

$$U = E = \frac{c^4 R_A}{2G} = \rho V, \qquad P = \frac{\rho}{3}, \qquad \boxed{dE = T\, dS - P\, dV.} \tag{7}$$

For flat FRW, $R_A = c/H$ and the expansion, acceleration, and conservation laws are

$$\boxed{H^2 = \frac{8\pi G}{c^2}P}, \qquad \boxed{\frac{2\ddot{a}}{ac^2} = -\kappa_E(P + p_m)}, \qquad \boxed{\dot{P} + H(3P + p_m) = 0}, \tag{8}$$

Here $H = \dot{a}/a$, $\kappa_E = 8\pi G/c^4$, ρ is total bulk energy density and p_m total bulk pressure, distinct from P; vacuum contributions are included consistently. For curved FRW, $R_A = c/\sqrt{H^2 + kc^2/a^2}$ and $1/R_A^2 = \kappa_E P$. These reorganize the Einstein–FRW equations.

Canonical basis and scope

With fixed clock, cuts, and the stated mixed boundary prescription, the scalar nonexpanding-null response ($\theta = \delta\theta = 0$) is [1]

$$\frac{\Theta_\kappa^{(0)}}{\Delta\tau} = \frac{Ac^2}{8\pi G}\,\delta\kappa = -\frac{Ac^4}{8\pi GL^2}\,\delta L = -F_L\,\delta L. \tag{9}$$

This selected response per prescribed time is realized by stationary Rindler, Schwarzschild and de Sitter. It is not the full action variation. Evolving FRW retains its separate equilibrium-screen calibration; no complete moving-boundary canonical theorem is asserted. Thermal input is unnecessary for F_L and P_L.

[1] Hopfmüller–Freidel, 1611.03096; Odak et al., 2309.03854; Parattu et al., 1303.1535.

[2] Cai–Kim, hep-th/0501055; Akbar–Cai, hep-th/0609128. Details and scope: the companion seven-page and full manuscripts.